

Topic 1

Project Management Concepts



What Is a Project?

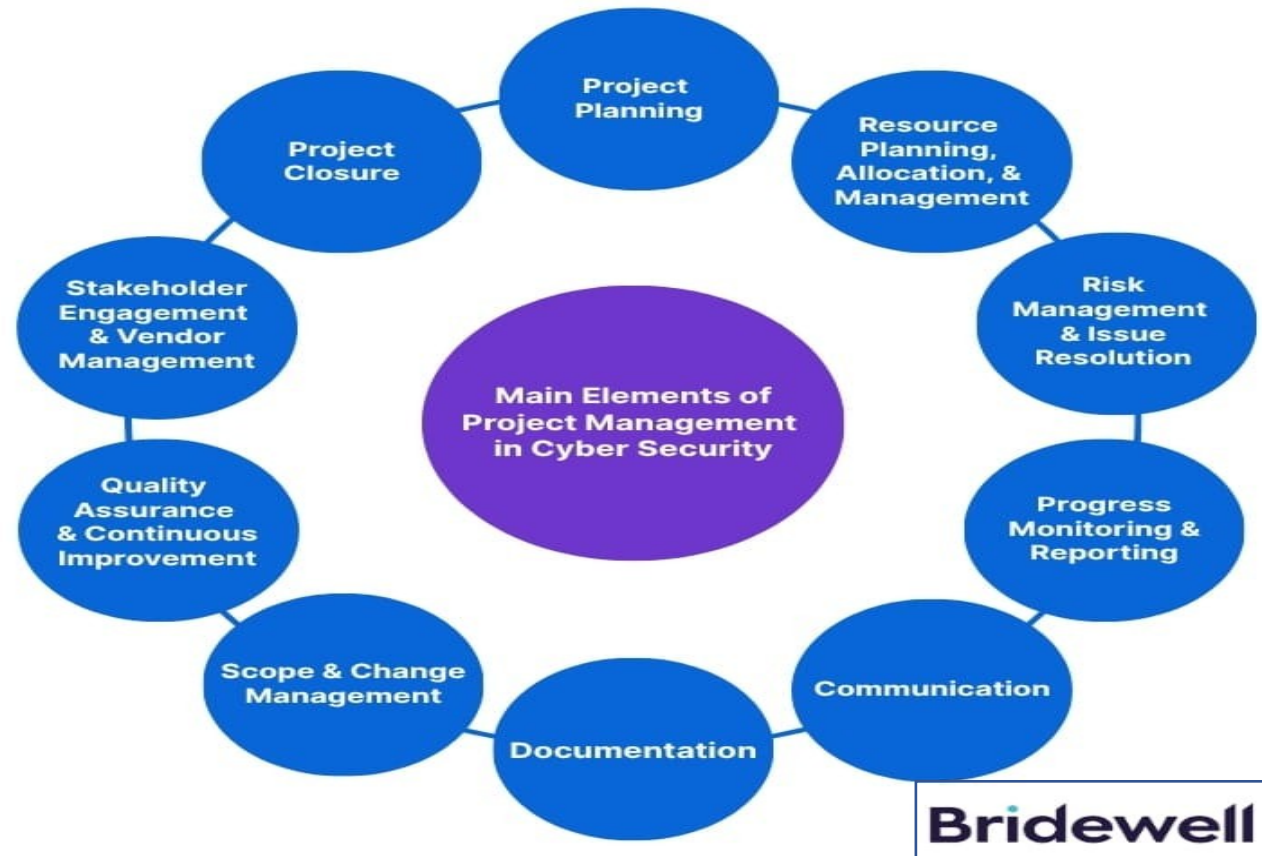
- ▶ A **project** is “a temporary endeavor undertaken to create a unique product, service, or result”
- ▶ Operations is work done to sustain the business
- ▶ Projects end when their objectives have been reached or the project has been terminated
- ▶ Projects can be large or small and take a short or long time to complete

Examples of IT Projects

- ▶ A team of students creates a smartphone application and sells it online
- ▶ A company develops a driverless car
- ▶ A small software development team adds a new feature to an internal software application for the finance department
- ▶ A college upgrades its technology infrastructure to provide wireless Internet access across the whole campus

Project Management in Cyber Security

Project management in cyber security is about more than just planning, executing, and overseeing tasks. It's about protecting an organisation's digital assets - including computers, data, networks, and systems - against ever-evolving threats.

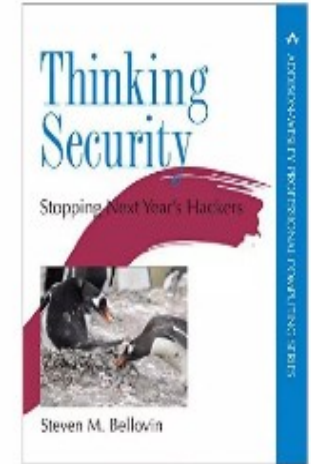


Security Concepts

- Security is the **new black**
- Security is an issue of technical debt
- Challenges
 - **How to Deliver "Secure"**
 - **How to Deliver "Securely"**
 - How to Deliver "Security"



- Confidentiality, Integrity, Availability, ...
- Terminology
 - Vulnerabilities
 - Threats
 - Risk
- Compliance != Security



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- The CIA Triad: Understanding Confidentiality, Integrity, and Availability as the core principles of information security.
- Key Terms: Vulnerability, Threat, Exploit, and Risk.
- Frameworks: Familiarity with industry standards such as the NIST Cybersecurity Framework (Identify, Protect, Detect, Respond, Recover).

Cybersecurity is a critical concern for businesses of all sizes and industries. Project management, in particular, requires a robust cybersecurity strategy to protect sensitive data and information throughout the project lifecycle. With cyber threats becoming more sophisticated and prevalent, project managers need to stay informed and proactive in implementing cybersecurity measures. This blog will explore the importance of cybersecurity in project management and provide valuable insights into best practices for safeguarding project data and assets.

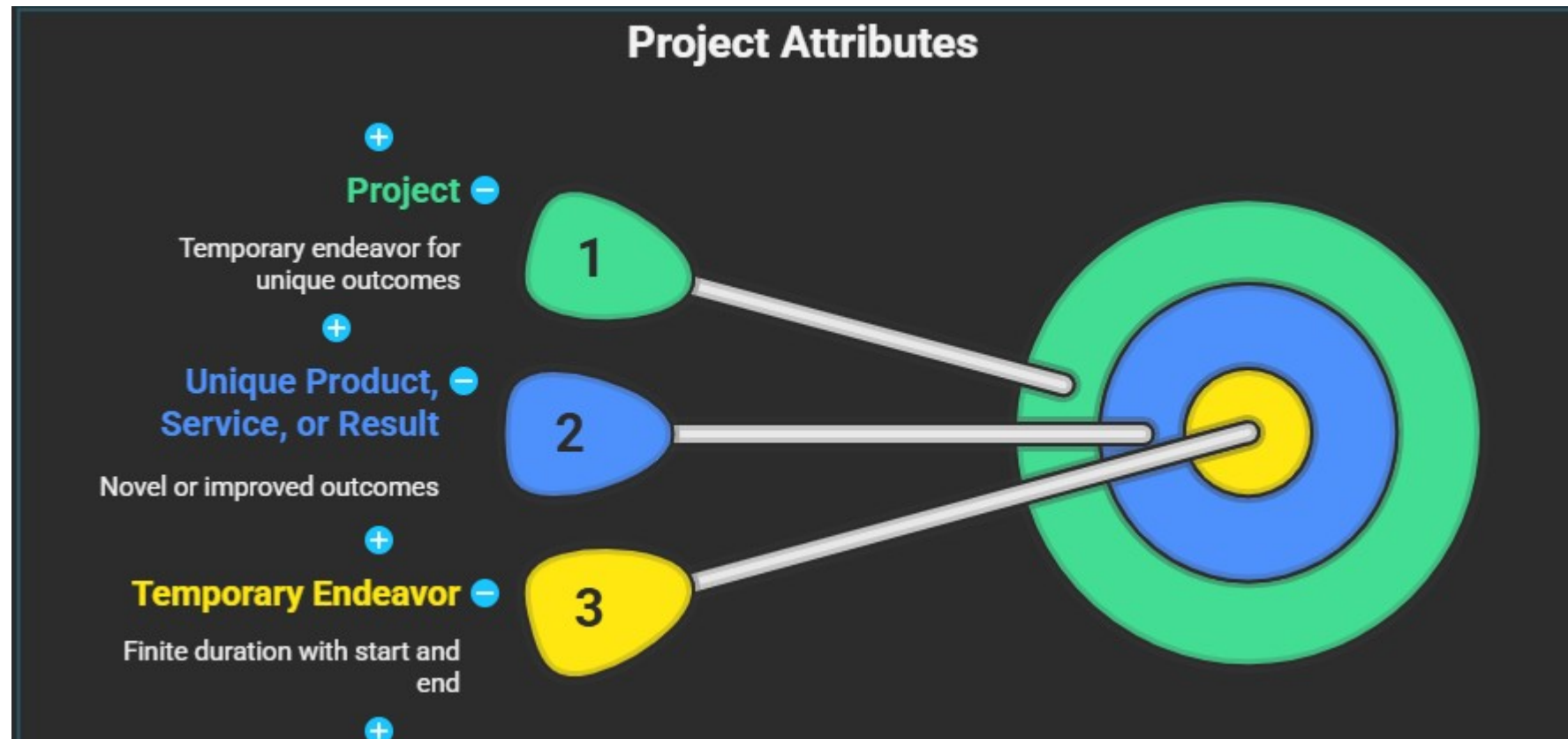
Importance Of Cybersecurity In Project Management



- 1. Prevention Of Data Breaches:** Cybersecurity measures in project management are crucial to prevent unauthorized access to sensitive project data, which could lead to data breaches and compromise of confidential information. By implementing robust cybersecurity protocols, project managers can minimize the risk of data breaches and protect the project's integrity.
- 2. Protection Of Intellectual Property:** Projects often involve valuable intellectual property and proprietary information that must be safeguarded from cyber threats. Cybersecurity measures like encryption, access control, and regular data backups can help project managers protect their intellectual property from cyberattacks and data theft.
- 3. Ensuring Business Continuity:** Cyberattacks can disrupt operations and jeopardize project timelines and deliverables. By investing in cybersecurity measures, project managers can ensure business continuity by mitigating the impact of cyber incidents and maintaining the smooth functioning of the project despite potential cyber threats.
- 4. Compliance With Regulations:** Many industries have specific regulations and compliance requirements related to data security and privacy. By integrating cybersecurity best practices into project management processes, project managers can ensure compliance with regulatory standards and avoid costly penalties or legal consequences resulting from data breaches or security incidents.
- 5. Building Stakeholder Trust:** Effective cybersecurity practices demonstrate a commitment to protecting project assets and stakeholders' interests, which can enhance trust and credibility with clients, partners, and other project stakeholders. By prioritizing cybersecurity in project management, project managers can strengthen stakeholder relationships and create a secure environment for collaboration and communication.

Attributes of a Project

A project is a temporary endeavor undertaken to create a unique product, service, or result. Understanding the key attributes of a project is crucial for effective management. These attributes distinguish projects from ongoing operational work.

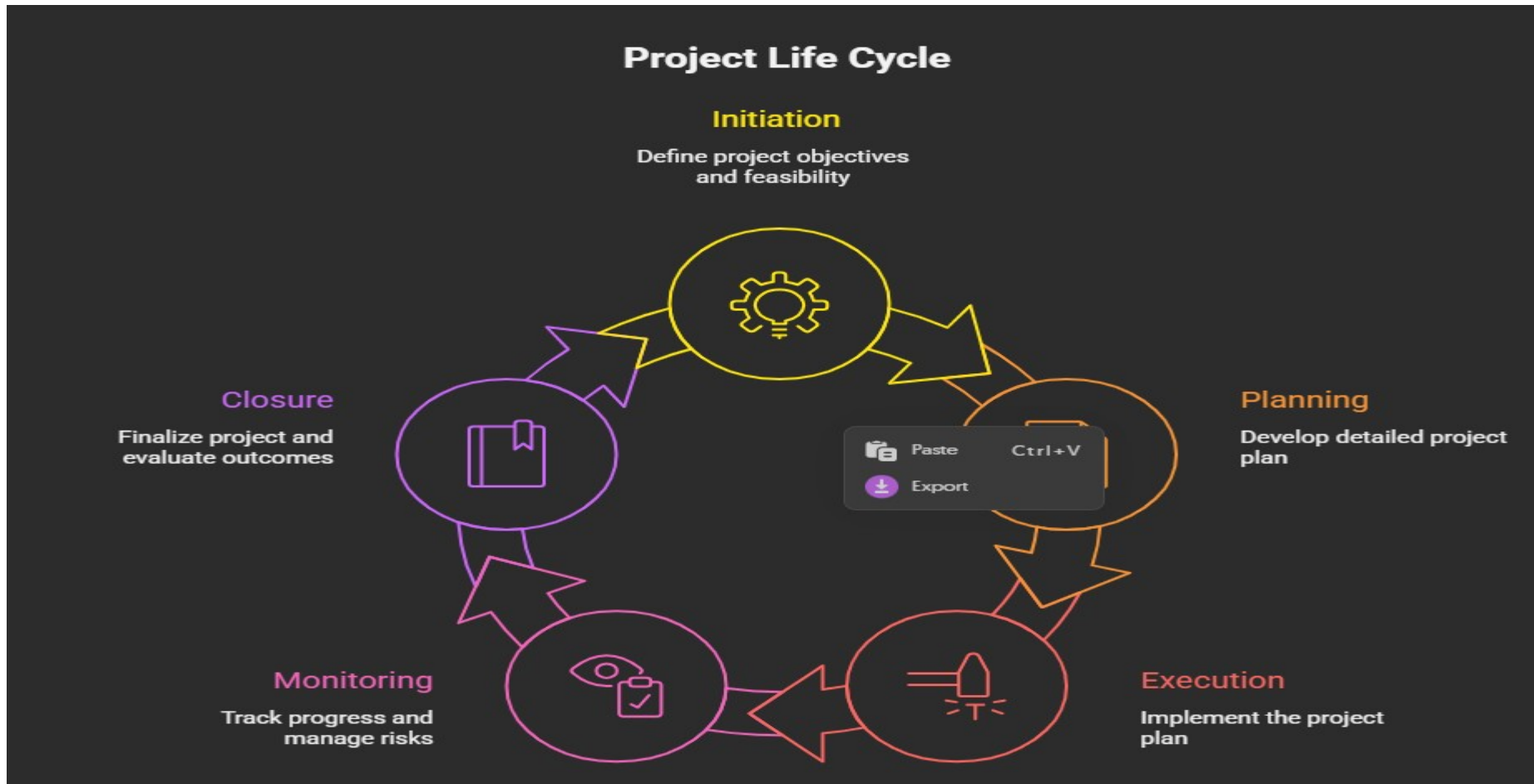


1.1 Attribute of a Project

- **Core Definition:** A project is a temporary endeavor undertaken to create a unique product, service, or result.
- **Temporary:** Every project has a definite **beginning and end**.
- **Unique Purpose:** It is not a routine operation; it results in a specific "deliverable."
- **Progressive Elaboration:** Projects are developed in steps and continue by increments.
- **Resource Consuming:** Requires various resources (human, financial, and hardware/software).
- **Primary Sponsor:** Usually has a main stakeholder or "Champion" who provides direction and funding.
- **Uncertainty:** Because every project is unique, it carries a degree of risk and unknowns.

1.2 Project Life Cycle

The project life cycle provides a framework for managing a project from its initiation to its closure. It defines the stages a project typically goes through and the activities performed in each stage. Different industries and organizations may use slightly different life cycle models, but the core principles remain the same. A common project life cycle includes the following phases:



1.2 Project Life Cycle

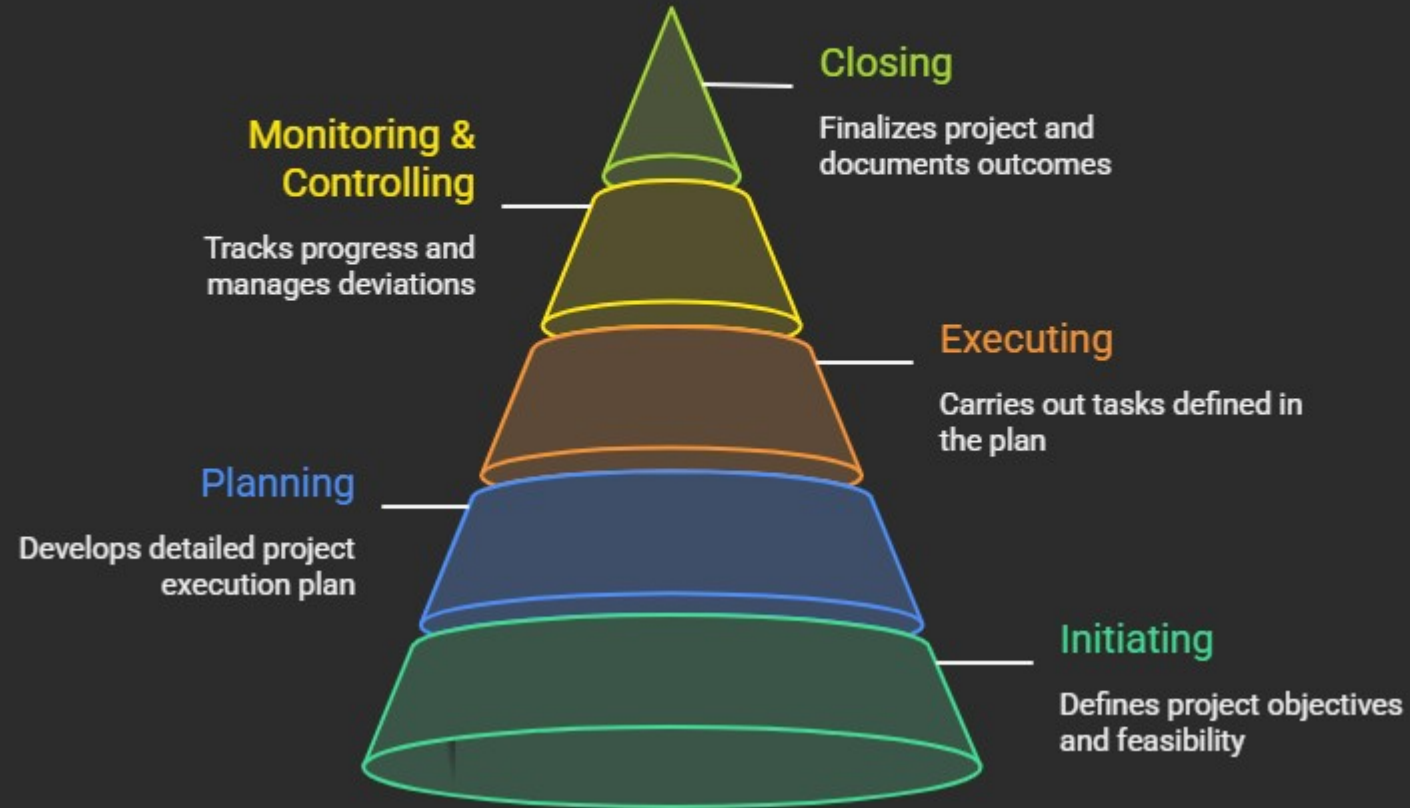
- **Initiation:** This is the starting phase of the project. It involves defining the project's objectives, scope, and feasibility.
- **Planning:** This phase involves developing a detailed plan for how the project will be executed.
- **Execution:** This phase involves carrying out the tasks defined in the project plan.
- **Monitoring and Controlling:** This phase involves tracking the project's progress, identifying any deviations from the plan, and taking corrective action. This phase runs concurrently with the execution phase.

1.3 The Project Management Process

The project management process encompasses the activities, methods, and tools used to manage a project effectively. It provides a structured approach to planning, executing, and controlling projects to achieve their objectives. The Project Management Institute (PMI) defines five process groups:

- **Starting the Project (Initiation):** Defining the project at a high level and obtaining authorization.
- **Organizing and Preparing (Planning):** Developing the roadmap (Scope, Schedule, Budget).
- **Carrying out the Work (Execution):** The phase where the "heavy lifting" happens and the product is built.
- **Monitoring and Controlling:** These processes involve tracking the project's progress, identifying any deviations from the plan, and taking corrective action. They include monitoring project performance, identifying variances, taking corrective action, managing risks, and reporting progress.
- **Closing the Project (Closure):** Formalizing acceptance and handing over the final product.

Project Management Process Pyramid



1.4 Benefits of Project Management

- ▶ Clear goals and priorities.
- ▶ Better resource utilization.
- ▶ Improved risk management.
- ▶ Higher quality outcomes.
- ▶ Enhanced stakeholder satisfaction.

Why do organizations invest in formal PM methodologies?

Benefit	Description
Efficiency	Provides a "roadmap" that avoids reinventing the wheel.
Customer Satisfaction	Projects completed on time and within budget lead to happy clients.
Improved Growth	Standardized processes allow the team to handle more complex projects.
Risk Mitigation	Potential issues are identified early through a "Red/Amber/Green" status approach.
Better Standing	Enhances the reputation of the IT department within the organization.

Benefits of Project Management

**Streamlines
the Workflow**



**Enhances
Stakeholder
Relationships**



**Anticipates
Problems
Early On**



**Increases
Accountability
Among Members**



**Keeps the
Team
Aligned**

